

01 PROFILE	Design Engineer taking AI-native interfaces from concept to production : knowledge-graph explorers, multi-agent pipeline UIs, and research instruments. I own both the interaction model and the production React/TypeScript behind it. A decade of hardware prototyping sits underneath the software: Apple Vision Pro labs, VR hardware, medical devices. AI-first builder: Claude Code and Cursor are daily practice.		
02 CORE SKILLS	BUILD	React · TypeScript · Python · FastAPI · Node.js · GraphQL · SSE & real-time state in data-intensive apps · design systems & component libraries · data visualization & knowledge graphs	
	DESIGN	Interaction & motion design · information hierarchy for data-dense, trustworthy AI interfaces · design research (surveys, depth interviews) · rapid prototyping, software & hardware · accessibility, reduced motion, performance budgets	
	AI-NATIVE	Claude Code & Cursor agentic workflows as core practice · LLM integration & multi-agent UX · prompt-to-production pipelines · making agent reasoning legible to users	
03 EXPERIENCE	Design Engineer & Product Lead Futurity Systems, Barcelona		Feb 2025 – Present
	FAST, AI-accelerated science & foresight platform. Lead frontend engineer and lead UX/UI designer: a data-dense research interface built for trust. GPU-accelerated knowledge-graph explorer that keeps ten-thousand-node graphs readable, React Flow analysis-workflow DAGs, multi-step LLM invention tools, plus a companion browser extension for one-action research capture. 400+ component production React/TypeScript codebase on a Chakra-based design system. Futurity Engine, multi-agent research engine. Built the UI concept prototype that established agent reasoning as a navigable graph, then the production frontend with the team. SSE streams every step of a five-phase agent pipeline into a live knowledge graph (25+ semantic node types) so the system stays legible while it works. Backend contributions in the FastAPI orchestrator. Futurescaper, futures-exploration tool. End-to-end ownership: solo rebuild from a colleague's browser-only proof-of-concept into a full product. The UI, a custom graph-layout system tuned for foresight data, and the backend owning AI orchestration. Futures Garden, EU initiative (2025). Concept and interaction design on a team project: an exhibit where visitors talk to LLM-powered archetypes ("digital souls") by tapping NFC-tagged phygital objects on a physical Orb. AI-native practice. Claude Code and Cursor in the loop for scaffolding, exploration, and refactoring across all product work. Including carlton.dev, designed and built end-to-end with AI agents as collaborators.		
	Hardware Associate Engineer Apple (Vision Pro)		2022 – 2023
	Lab buildout & operations. Planned and built two prototyping labs for Vision Pro teams (floorplan, budget, equipment: 3D printing, laser cutting, test gear): the Product Design team's on-site lab and the shared Hardware/Firmware lab. Ran them day to day alongside product-design, hardware, and firmware engineers. Managed a small team and completed Apple management training.		
	Product Development Lab Manager Pensar Development, Seattle		2016 – 2022
	Prototyping lab. Ran the lab for a product-development consultancy: prototyping across consumer electronics, VR hardware, and medical devices, all under NDA. LabVIEW-automated test fixtures (PXI), CNC machining, SLS/FDM printing, wire harnessing, PCB rework. Production leadership. Planned and led a 500-unit hardware assembly program at a client factory: 8-person team, station design, training, and the test program. The pivot to software. Built a full-stack lab-equipment tracking portal with QR-code check-out (React, Node.js, GraphQL) and shipped it to the whole engineering team.		
	Earlier: Senior PCB Repair Technician, FUJIFILM SonoSite (2015–16): component-level repair of ultrasound systems · Robotics Technician, Hydromax USA (2014–15): robotic inspection systems for water infrastructure.		
04 EDUCATION & RESEARCH	Master's, Interaction Design Harbour.Space University, Barcelona		2023 – 2025
	Graduate research with Campus AI (2024). Qualitative study of mid-career career-shifters on an LLM learning platform: 42-respondent survey, eight 90-minute depth interviews, 1,117 coded moments. Synthesized into product strategy centered on capability growth over engagement (team of four).		
05 PORTFOLIO	carlton.dev Interactive case studies with live, working demos and honest ownership breakdowns. Zero-framework JS, performance-budgeted, reduced-motion aware. Typography, deliberate motion, considered interaction as craft.		